



Use of Quartz Cementation Modelling to Constrain Burial Histories of the Zulia Oriental Region, Western Venezuela

Pérez, R.¹; Chatellier, J-Y.¹; Lander, R.², 1998

¹ PDVSA-Intevep. Los Teques, Edo. Miranda, Venezuela.

² Geologica, PO Box 8034 4003 Stavanger, Norway.

Resumen

The purpose of this paper is to show how two different paleothermal indicators can be used in concert to increase understanding of the thermal and burial history of a basin. This investigation was prompted by the occurrence of large differences in apparent thermal histories in two neighboring wells in the Zulia Oriental region of Eastern Venezuela. Our simulations of the abundance of quartz cement in sandstones helped to confirm previous thermal reconstructions that were calibrated using vitrinite reflectance data. Although both of these paleothermal indicators are highly sensitive to temperature, vitrinite reflectance values tend to be dominantly controlled by the maximum temperature, whereas quartz cementation tends to be more sensitive to the elapsed time at high temperatures. The application of the two thermal indicators make it possible to constrain the overall thermal history to a high precision as well as distinguish different tectonic provinces in the study area.